



Functional Requirements & Database Schema

Specification v1.0 · Bangladesh advocate practice-management app · Prepared for backend build & review.

Scope · Single-advocate diary, multi-tenant

Recommended stack · Supabase (Postgres + Auth + Storage)

- MUST — MVP
- SHOULD — v1.1
- COULD — later

1 · Functional Requirements

A · Authentication & Accounts

FR-A1	An advocate can register with full name, mobile number, email and password; email must be unique.	MUST
FR-A2	Login by email + password, followed by a 6-digit OTP (SMS/email) verification step.	MUST
FR-A3	Password reset / "forgot password" via email or mobile OTP.	MUST
FR-A4	Persistent session ("remember this device") with secure token; explicit logout.	SHOULD
FR-A5	Each account is an isolated tenant — a user can never read or write another advocate's data.	MUST
FR-A6	Optional junior/clerk sub-users with scoped access under one chamber.	COULD

B · Advocate Profile

FR-B1	Upload / change a profile photo; diary header shows the photo and "{First}'s Diary".	MUST
FR-B2	Edit name, chamber/firm, Bar enrolment no., contact details.	MUST
FR-B3	Light / night-mode preference saved per user.	SHOULD
FR-B4	Default language (English / Bangla) for client messages.	SHOULD

C · Case Management

FR-C1	Create, view, edit and delete a case (case no., type, court, stage, summary, notes).	MUST
FR-C2	Case types: Civil, Criminal, Family, Writ, Revision, Appeal, etc. (extensible list).	MUST
FR-C3	Inline edit panel updates fields; changes persist on save with confirmation.	MUST
FR-C4	A read-only Case File view shows order-sheet history, payments and client details.	MUST
FR-C5	Per-case private "to-do / DO" note, editable by the advocate.	MUST
FR-C6	Court's order / proceeding text is recorded per hearing (shown as court-fixed).	MUST
FR-C7	Soft-delete with undo window; hard-delete after confirmation.	SHOULD
FR-C8	Tag / link related cases (e.g. a revision connected to a suit).	COULD

D · Parties & Representation

FR-D1	Each case has ordered plaintiff(s)/petitioner(s) and defendant(s)/respondent(s) with numbers (A1, B1...).	MUST
FR-D2	Mark which party/parties are "my client(s)"; my-client labels show the party number.	MUST
FR-D3	Opposition lawyer(s) recorded with name + contact, and which numbered party they appear for.	MUST
FR-D4	Support multiple opposition lawyers, each mapped to one or more defendants.	SHOULD
FR-D5	Reuse a client across multiple cases; client profile aggregates their matters.	MUST

E · Scheduling & Forwarding

FR-E1	Record previous date, today's fee and a forward-to-next-date in one action.	MUST
FR-E2	"Forward & keep record" moves the matter to the new date and leaves a dated appearance record on the old date.	MUST
FR-E3	A case with no next date is flagged and surfaced for follow-up.	MUST
FR-E4	Each forward appends an entry to the order-sheet / hearing history.	MUST
FR-E5	Detect/ warn on a date clash (two hearings same day) — informational.	COULD

F · Calendar & Cause List

FR-F1	Month calendar with a per-day hearing count badge.	MUST
FR-F2	Selecting a date shows that day's cause list; "Today" and clear-date shortcuts.	MUST
FR-F3	At-a-glance counters: today, next 7 days, active matters.	MUST
FR-F4	All-Cases page: full list searchable by mobile no, case no, party, court or type.	MUST
FR-F5	Printable / exportable daily cause list.	SHOULD

G · Reminders & Notifications

FR-G1	Personal day reminders (to-dos) per calendar date, add/edit/complete/delete.	MUST
FR-G2	Automatic recurring reminder (every ~2 days) for any matter with no next date.	MUST
FR-G3	Pre-hearing reminder N days before a next date.	SHOULD
FR-G4	Delivery channels: in-app, push, email; optionally SMS/WhatsApp.	SHOULD
FR-G5	Per-user notification preferences and quiet hours.	COULD

H · Payments & Fees

FR-H1	Record fee receipts per case, typed as honorarium/retainer or task/step fee.	MUST
FR-H2	Case File shows a payment diagram splitting honorarium vs task fees with totals.	MUST
FR-H3	Currency BDT (₳); amount, date and label per entry.	MUST
FR-H4	Outstanding / agreed-fee tracking and balance due.	SHOULD
FR-H5	Export a client fee statement (PDF).	COULD

I · Documents & Files

FR-I1	Upload JPG/PNG/PDF/DOC files against a case; list, open, print, remove.	MUST
FR-I2	Files stored in object storage; access via short-lived signed URLs.	MUST
FR-I3	File size/type validation and virus scanning on upload.	SHOULD
FR-I4	Capture from phone camera (scan document).	COULD

J · Client Communication

FR-J1	Generate a client update message (English/Bangla) from the latest order + next date.	MUST
FR-J2	Send via WhatsApp deep-link or copy to clipboard.	MUST
FR-J3	Log which updates were sent and when.	SHOULD
FR-J4	Templated, official WhatsApp Business API delivery.	COULD

K · Search

FR-K1	Global search across case no, party names, mobile numbers, court and type.	MUST
FR-K2	Filter by date, case type and "needs next date".	SHOULD
FR-K3	Full-text search inside notes / order sheets.	COULD

L · Security, Isolation & Audit

FR-L1	Row-Level Security enforces tenant isolation at the database.	MUST
FR-L2	All traffic over HTTPS; secrets server-side only.	MUST
FR-L3	Encrypt files and database at rest.	MUST
FR-L4	Audit log of create/update/delete on cases, payments and files.	SHOULD
FR-L5	Per-user data export and account deletion (data-protection).	SHOULD
FR-L6	Automated encrypted backups with tested restore.	MUST

M · Non-functional

FR-M1	Mobile-first responsive UI; primary tap targets ≥ 44 px.	MUST
FR-M2	Works on low-end Android browsers; usable on slow networks.	MUST
FR-M3	Installable as PWA (offline read of cached cases).	SHOULD
FR-M4	Packageable as an Android APK (Capacitor / TWA wrapper).	SHOULD
FR-M5	Bangla + English UI strings (i18n-ready).	SHOULD

2 • Database Schema

PostgreSQL. Every tenant-owned row carries **advocate_id**; Row-Level Security restricts each advocate to their own rows (true per-user isolation). Primary keys are **uuid**; timestamps are **timestamptz** with **created_at** / **updated_at**.

advocates one row per registered lawyer (tenant root)		
id	uuid PK	matches auth user id
full_name	text	e.g. "Adv. Shaon Rahman"
email	citext UNIQUE	login identity
phone	text	mobile number
bar_enrolment_no	text	optional
photo_path	text	storage key for profile picture
theme	text	'light' 'dark'
default_lang	text	'en' 'bn'

clients people/companies the advocate represents		
id	uuid PK	
advocate_id	uuid FK → advocates	tenant owner
name	text	
role	text	e.g. Plaintiff / Petitioner
phone	text	searchable
address	text	
client_since	date	

cases a matter / suit		
id	uuid PK	
advocate_id	uuid FK → advocates	tenant owner
client_id	uuid FK → clients	primary client (nullable)
case_no	text	e.g. "Title Suit 184/2024"
type	text	Civil/Criminal/Family/Writ...
court	text	full court name
court_short	text	abbreviated
stage	text	current stage label
stage_type	text	evidence/hearing/argument/pending
prev_date	date	last hearing
next_date	date	next hearing (null ⇒ reminder)
court_order	text	latest order / proceeding
todo	text	private "DO" note
notes	text	special notes
summary	text	matter summary

case_parties ordered parties on each side		
id	uuid PK	
case_id	uuid FK → cases	
side	text	'plaintiff' 'defendant'
position	int	1,2,3 → A1/B2...
name	text	
is_mine	boolean	true ⇒ my client
client_id	uuid FK → clients	nullable link
phone	text	

opposing_counsel opposition lawyers + who they appear for		
id	uuid PK	
case_id	uuid FK → cases	
name	text	
phone	text	
represents	int[]	party positions on the other side

hearings order-sheet / appearance history

id	uuid PK	
case_id	uuid FK → cases	
date	date	hearing date
kind	text	filing/order/evidence/hearing/disposal
title	text	short headline
note	text	what happened
forwarded_to	date	next date taken (nullable)

payments fees received from clients

id	uuid PK	
case_id	uuid FK → cases	
advocate_id	uuid FK → advocates	
type	text	'honorarium' 'task'
label	text	description
amount	numeric(12,2)	BDT
paid_on	date	

documents uploaded case files

id	uuid PK	
case_id	uuid FK → cases	
file_name	text	
storage_path	text	object-storage key
mime_type	text	
size_bytes	bigint	
uploaded_at	timestampz	

reminders personal day to-dos

id	uuid PK	
advocate_id	uuid FK → advocates	
on_date	date	the day it belongs to
text	text	
done	boolean	

notifications_log sent reminders / client updates		
id	uuid PK	
advocate_id	uuid FK → advocates	
case_id	uuid FK → cases	nullable
channel	text	inapp/push/email/sms/whatsapp
kind	text	blank-date / pre-hearing / client-update
sent_at	timestamptz	

3 • Relationships & Rules

- ▶ advocates 1—∞ clients, cases, payments, reminders. Deleting an advocate cascades to all owned data.
 - ▶ cases 1—∞ case_parties, opposing_counsel, hearings, payments, documents (ON DELETE CASCADE).
 - ▶ clients 1—∞ cases; a client may appear on many cases, a case has one primary client + many parties.
 - ▶ opposing_counsel.represents holds the numbered defendant/plaintiff positions a counsel appears for (D3/D4).
 - ▶ A case is "needs next date" when next_date IS NULL — drives the recurring reminder job (FR-G2).
 - ▶ RLS policy on every tenant table: advocate_id = auth.uid() (cases reached via their advocate_id; children via their parent's advocate_id).
 - ▶ Indexes: cases(advocate_id, next_date), case_parties(case_id), payments(case_id), and a search index over case_no / party name / phone.
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